

PROCESSING OF TEMPORAL INFORMATION IN THE BRAIN

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INTRODUCTION AND SCOPE OF THE REVIEW

Nervous systems can resolve microsecond time differences. These time differences would seem too small for single neurons to resolve, because neural events occur on a millisecond, rather than a microsecond, time scale. Accurate coding of temporal information is widespread, however, and many theories of sensory biology depend upon the detection of signals that are correlated in time. This review summarizes studies of time coding in the central nervous system (CNS) to show how the sensitivity to time differences arises. The first part summarizes behavioral sensitivity to time differences. The second describes specializations for the coding of temporal information in the CNS. The last describes the neural circuits underlying detection of time differences.

The review concentrates on three specialists—weakly electric fish, barn owls, and echolocating bats—because they have well developed abilities to detect time differences. Time coding in other animals is also discussed where possible, although a complete description of the temporal coding of complex stimuli is beyond the scope of this review. Analysis of how animals process temporal information has uncovered many common principles; despite their different neural substrates, time-coding systems implement similar algorithms for the encoding of temporal information. An algorithm refers here to steps and procedures in the processing of a signal (Konishi 1991). In time-coding systems, the timing of the stimulus is coded by phase-locked spikes, timing information is processed in a dedicated pathway in parallel with other stimulus variables, and the time-coding cells have morphological and physiological features suited to their function. Further-

more, time-coding circuits show increasing specializations for the extraction of relevant temporal features needed to measure time differences. The circuits that measure time differences in different animals also employ similar algorithms. All circuits that detect time differences depend on some form of delay lines and coincidence detectors. The coincidence detectors are neurons that respond maximally when they receive simultaneous inputs. This can occur when the time difference is compensated for by an equal and opposite delay of the inputs. The coincidence detector circuits use temporal information to compute the new variable of time difference.

BEHAVIOR AND DETECTION OF TEMPORAL INFORMATION

In an influential series of papers, Simmons (1973, 1979) focused attention on the problem of detecting small time differences. He showed that the bat *Eptesicus fuscus* could detect changes as small as 500 ns in the arrival time of jittered sonar echos. There followed several clear demonstrations of sensitivity to small time differences that included discrimination of phase differences between signals on different parts of the body surface in electric fish (Carr et al 1986a; Rose & Heiligenberg 1985) and detection of binaural phase differences for sound localization in owls, bats, and many other mammals (Heffner & Heffner 1992; Masters et al 1985; Moiseff & Konishi 1981). The behavioral studies outlined below identify which features of the stimulus the animal attends to, and describe each animal's ability to detect time differences.

Weakly electric fish generate an electric field around their body by discharging an electric organ. The field is detected by arrays of electroreceptors distributed over the body surface. Phase differences between signals on different parts of the body surface are essential cues for the electrolocation of objects and for communication with conspecifics, and the fish have a well-developed system for encoding and processing phase information (for review, see Heiligenberg 1991). The electric fish *Eigenmannia* produces a quasisinusoidal electric organ discharge in the range of 400–600 Hz. When these fish encounter neighbors with similar discharge frequencies, the electric signals add in the water and produce a beating signal that interferes with each animal's ability to electrolocate, because the small disturbances caused by the presence of a prey item are masked by the large changes in amplitude and phase caused by the interfering signals. Electric fish have developed a response to avoid jamming, whereby if a fish meets a conspecific with a similar frequency, each fish shifts the frequency of its electric organ discharge so as to increase the frequency difference between them (Bullock et al 1972). The fish determines whether

its neighbor has a higher or lower frequency than its own by evaluating the fluctuations in amplitude and phase of the beating signal on different parts of its body surface (Heiligenberg et al 1978). Correct performance of the jamming avoidance response, therefore, requires evaluation of differential phase, and this behavior provides an assay for the minimum phase difference that can be perceived by the fish (Carr et al 1986a; Rose & Heiligenberg 1985).

Behavioral experiments have demonstrated that these electric fish detect phase differences in the microsecond range (Carr et al 1986a). The fish is presented with weaker and weaker jamming stimuli that produce smaller and smaller phase differences between different parts of the body surface. Correct performance of jamming avoidance responses to these weak jamming stimuli can be used to show that the fish can detect phase differences as small as 400 ns between different parts of the body surface. Physiological recordings, however, show that the phase-locked responses of afferents are too jittery to permit such fine temporal resolution (see next section). How can the fish, using only the information gathered from any single afferent, reliably perform a correct jamming avoidance response when the maximal phase difference is less than 1 μ s? Hyperacuity is a perceptual term to describe the phenomenon whereby sensory thresholds are lower than expected from the properties of individual receptors (see Altes 1989). Rose & Heiligenberg (1985) showed that this temporal hyperacuity must result from the convergence within the CNS of parallel phase-coding channels from sufficiently large areas of the body surface, because the ability to detect the small phase differences diminished when smaller numbers of receptors were stimulated.

The barn owl's ability to detect time differences is acute. These owls can catch mice in total darkness on the basis of auditory cues alone (Konishi 1973; Payne 1971). A sound coming from one side of their body reaches one ear before the other, and the owl translates this interaural time difference into location in azimuth (the horizontal plane). The owl actually derives interaural time differences from the interaural phase differences present in the auditory stimulus. Behavioral experiments show that the owl uses the phase differences between the two ears, rather than stimulus onset time to localize sound, because if sound is presented through earphones, and phase delayed in one ear with respect to the other ear, the owl will turn its head in the direction predicted by the phase difference (Moiseff & Konishi 1981). Barn owls are very accurate (Konishi 1973); Knudsen & Konishi (1979) determined the average error made in free field sound localization by having the owl turn its head toward a sound in the dark and found that the owl's average error in turning its head to a sound was 1.5° (about 1.5 μ s interaural time difference).

Most animals use interaural phase differences to localize sound (Fay 1988; Heffner & Heffner 1992). Animals with large heads have much larger time differences available to them; conversely, animals with small heads have to achieve much greater resolution of binaural time differences than a large animal to obtain the same degree of accuracy. For example, barn owls and humans have very similar abilities to localize sound; psychophysical studies have shown that human subjects can localize a frontal tone with an accuracy of about 1° (Mills 1972). The human discrimination task is easier than the barn owl's, because the human head is larger. Sound localization ability is not always well correlated with head size, however. Instead, it is highly correlated with directing the attention of the other senses to the sound source. Animals with narrow fields of best visual acuity require accurate sound localization to direct their gaze, whereas the echolocating mammals, bats, and dolphins use sound localization to direct their biosonar pulses (Heffner & Heffner 1992).

Echolocating microchiropteran bats emit biosonar pulses and listen to the returning echos. They determine target distance by measuring the time delay between the outgoing pulse and the returning echo (Simmons 1973, 1979). A bat can be trained to respond to one of two simultaneously presented stimuli and can detect changes as small as 500 ns in the arrival time of sonar echos when these changes appear as jitter or alterations in arrival time from one echo to the next (Moss & Schitzler 1989; Simmons 1979). Simmons et al's (1990) most recent findings suggest that trained *Eptesicus fuscus* can detect jittered targets with a delay acuity of 10 ns. The psychophysical function relating the bat's performance to the magnitude of the jitter corresponds to a cross-correlation function between the emitted sonar signals and the echos, and the data are consistent with the hypothesis that the bat acts as an ideal sonar receiver that perceives the phase or period of the signals (Moss & Simmons 1992). The bat obtains an acoustic image of the target that is apparently derived from time domain or periodicity information processing by the nervous system (Simmons et al 1990). The neural basis of this nanosecond delay sensitivity is hard to imagine.

ENCODING OF TEMPORAL INFORMATION

Discrimination of small time differences requires accurate transduction and processing of the original stimulus. This section addresses how time coding arises in the periphery, and how this timing information is preserved and improved in the CNS. Furthermore, behavioral hyperacuity is assumed to involve averaging in large neural assemblies, in addition to

accurate responses from single neurons. Both strategies may be found in systems that have achieved fine temporal resolution.

The Electrosensory System

The electric organ discharge is detected by sensory hair cells, termed electroreceptors. Two kinds of electroreceptors encode the phase or amplitude of the electric signal (Scheich et al 1973; Szabo 1965). Nerve fibers that innervate the phase-coding type of electroreceptor fire one spike on each cycle of the stimulus, phase-locked with little jitter to the zero-crossing of the stimulus. The degree of phase-locking to the stimulus is often quantified by using a measure, termed vector strength (Goldberg & Brown 1969). The distribution of phase angles in the period histogram is measured: If most spikes fall within a small range of phase angles, the vector strength is high (near 1); if spikes are evenly distributed, the vector strength is near 0.

Phase-locking must originate at the level of the receptor, although there are few studies of hair cells that address this issue. Fortunately, in mormyrid electric fish, the Knollenorgan electroreceptor produces a phase-locked spike in response to the electric stimulus (Bennett 1970). These electroreceptors appear to be the only vertebrate hair cells that generate a spike. At the best frequency, about 1.1 kHz, the vector strength is high, about 0.94; it diminishes to 0.2 at 10 kHz (measured from Figure 6 in Hopkins 1986). These frequency-dependent decreases in vector strength have also been found in the auditory system, and they have been interpreted as showing that the neurons ability to phase-lock decreases with frequency (see Weiss & Rose 1988).

In both mormyrid and gymnotiform electric fish, primary afferents convey phase-locked spikes to the lateral line lobe of the medulla, where phase and amplitude coding primary afferents terminate on different cell types (Bell et al 1989; Maler et al 1981; Von der Emde & Bleckmann 1991). Thus, the segregation of phase and amplitude receptors in the skin is reinforced by the central connections formed in the medulla. The separation of phase and amplitude information into two parallel channels is a common feature of all time-coding systems. In electric fish, the two channels not only have separate connections, but also distinct morphology. Phase-coding terminals are specialized for maintenance of phase-locked spikes; the afferents form large club terminals on large spherical cells that have few or no dendrites and thick axons. The spherical cells relay timing information directly to the midbrain torus. These connections are responsible for the fish's ability to detect small time differences, and are discussed below.

The accuracy of phase-coding improves with the progression from recep-

tors to primary afferents to spherical cells to giant cells in the midbrain torus (Figure 1) (Carr et al 1986a). The accuracy of the phase-coding was determined by measuring spikes from phase coders in the medulla and in the midbrain torus. The jitter of these spikes, defined as the standard deviation of the response time to the stimulus, decreased threefold with the progression from medulla to midbrain. The basis for this improvement of accuracy may lie in the convergence from afferents to higher level neurons. Afferents converge upon spherical cells, which converge on giant cells in the torus (Figure 1). Each torus cell receives inputs from between 9 and 16 phase-coding afferents. If giant cells were to average these inputs, they should in theory reduce the jitter by $\sqrt{N}$ of the number of inputs, or 3–4 times. This predicted improvement in jitter is close to the measured decrease in jitter from 30 μ s in the medulla to 10 μ s in the midbrain. The accuracy of even the best single neurons in these first stations of the time-coding pathway does not match that of the behavior, however (Carr et al 1986a; Rose & Heiligenberg 1985). Electric fish can resolve temporal disparities in the submicrosecond range, a temporal resolution far superior to that observed in the primary electrosensory afferents. This hyperacuity must result from network processing, and is discussed in the next section.

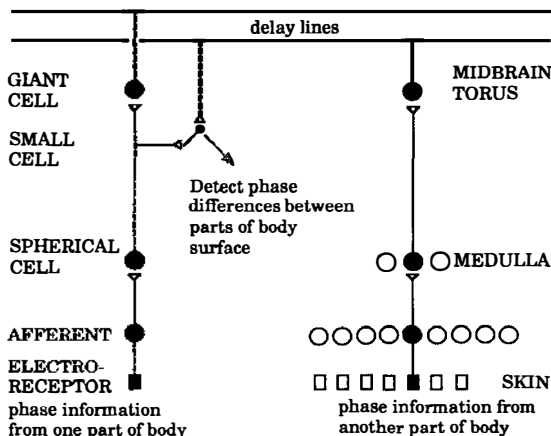


Figure 1 Schematic circuit in the electric fish midbrain for computation of phase differences between signals on any two parts of the body surface. Phase-coding electroreceptors converge on spherical cells in the medulla, which in turn converge in a topographic projection onto giant cell bodies and small cell dendrites in the torus. Giant cells relay the phase-locked signal all over the torus, with their terminals synapsing on the cell bodies of small cells. Small cells can, therefore, compare phase information from different parts of the body surface. Open circles are used to diagram the approximate convergence between different levels. (Modified from Carr et al 1986b and Heiligenberg 1991.)

The Auditory System

The behavioral experiments described in the previous section have shown that most animals use interaural phase differences to localize sound. Evidence of how they do so comes from experiments on the encoding and processing of phase information in the central auditory system. Time-coding systems appear to have changed little during amniote phylogeny, and although barn owls are useful models of temporal coding in the auditory system, studies on time coding in other birds, reptiles, amphibians, and mammals are equally instructive.

Recordings from auditory nerve fibers showed that spikes have a statistical tendency to phase-lock to the waveform of the acoustic stimulus (Kiang et al 1965). Spikes occur most frequently at a particular phase of the tone, although not necessarily in every tonal cycle. Thus, the discharge pattern of a cochlear nerve fiber can encode the phase of a tone with a frequency above 1000 Hz, even though the average discharge rate is low. The general assumption is that the modulating signal at the spike generator in the auditory nerve arises from components of the hair cell receptor potential, via the chemical synapse between the hair cell and the primary afferent. This timing information is degraded for high frequency sounds, presumably because of low pass filter effects in the hair cell (Kidd & Weiss 1990).

Cochlea hair cells encode and transmit both phase and amplitude information to the auditory nerve. Auditory nerve fibers phase-lock to the auditory stimulus and encode amplitude by increases in spike rate. Thus, there can be no predisposition toward coding for either amplitude or phase in the periphery, unlike electric fish. Nevertheless, the same parallel processing of phase and amplitude information that characterizes the electrosensory system is also found in the auditory system. The segregation does not begin at the level of the receptors, but differences in auditory nerve terminals may achieve a similar functional segregation in the CNS. To begin with the avian pattern, auditory nerve afferents enter the brain and then divide into two. One branch ramifies in the dendritic field of the cochlear nucleus angularis, which codes for changes in amplitude, whereas the other branch terminates in the cochlear nucleus magnocellularis, which codes for phase (Takahashi et al 1984). The terminal in the nucleus magnocellularis forms a specialized ending, termed an endbulb of Held (Brawer & Morest 1974). The endbulb synapse conveys the phase-locked discharge of the auditory nerve fibers to their postsynaptic targets in the nucleus magnocellularis. Thus, the synaptic specializations in the auditory nerve accomplish the same goal as the receptor specialization in electric fish. The endbulb is specialized to preserve phase-locked signals. Each endbulb has

multiple sites of synaptic contact on the soma to provide the substrate for the preservation of the phase-locked spikes between the auditory nerve fibers and the neurons of the nucleus magnocellularis. The endbulb is a secure and effective connection; physiological measures show that phase-locking is the same or better in the neurons of the nucleus magnocellularis than in the eighth nerve, whereas it is lost in the projection to the amplitude coding nucleus angularis (Sullivan & Konishi 1984).

Phase-locked spikes encode the timing of the stimulus, and the CNS uses this code for the measurement of time disparities. Phase information is preserved and improved, and interaural phase differences are detected, in a circuit composed of the auditory nerve, the cochlear nucleus magnocellularis, and the nucleus laminaris (Figure 2). Many of the features of this circuit may represent specializations for the encoding of timing information. The neurons of the nucleus magnocellularis are morphologically and physiologically specialized for the encoding of temporal information. They have large round cell bodies, a thick axon, and few medium length dendrites (Jhaveri & Morest 1982). In the owl, magnocellular neurons in the high best frequency regions of the nucleus have fewer dendrites than the neurons in the low best frequency region (Carr & Boudreau 1992). Similar changes in dendritic length are found in the nucleus laminaris in the chicken (Smith & Rubel 1979). These reductions in dendritic area might decrease the membrane time constant and improve the speed and accuracy of the phase-locked response to synaptic inputs.

The nucleus magnocellularis relays phase-locked spikes in a bilateral projection to the nucleus laminaris (Figure 2). There is a further improvement in phase-locking between the nucleus magnocellularis and the nucleus laminaris (Carr & Konishi 1990). The observed increase in phase-locking between magnocellular and laminaris neurons may be caused in part by the massive convergence of magnocellular afferents onto laminaris neurons. About 100 inputs from each magnocellular nucleus converge on each laminaris neuron. This convergence may serve two interdependent functions in the owl: It improves the accuracy of the phase-locking to the stimulus, as is the case with converging inputs in electric fish; and, it may produce a cycle-by-cycle representation of the stimulus in the postsynaptic cell and provide a substrate for coincidence detection of phase differences (see next section).

The circuits that encode temporal information in the mammalian auditory system are similar to those in birds, and the projection from auditory nerve fibers to spherical bushy cells in the anteroventral cochlear nucleus is homologous to the projection described in birds and reptiles (Boord 1968; Carr 1992; Ryugo & Fekete 1984; Spzir et al 1990). Mammalian bushy cells are morphologically and physiologically similar to magno-

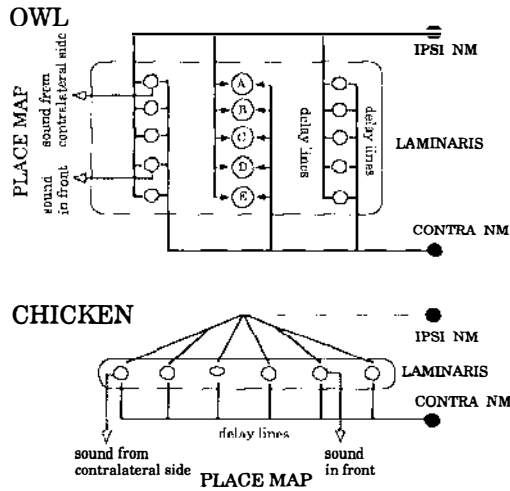


Figure 2 Jeffress model and schematic of brainstem auditory circuits for detection of interaural time differences in the barn owl and the chicken. Owl: Axons from the ipsilateral cochlear nucleus magnocellularis (IPSI NM) divide and enter the nucleus laminaris at several points along the dorsal surface. These axons act as delay lines within laminaris, interdigitating with inputs from the contralateral cochlear nucleus magnocellularis (CONTRA NM). In the center, the owl circuit has been modified to show the principles of the Jeffress model. Each binaural coincidence detector (A–E) fires maximally when inputs from the two sides arrive simultaneously. This can only occur when the interaural phase differences are compensated for by an equal and opposite delay. For example, neuron A fires maximally when sound reaches the contralateral ear first and is delayed by the long path from the contralateral ear so as to arrive simultaneously with the input from the ipsilateral ear. Thus, this array forms a map of interaural time difference in the dorso-ventral dimension of the nucleus. In the owl, sound from the front is mapped towards the ventral surface of the nucleus, and each nucleus appears to contain place maps of the contralateral and part of the ipsilateral hemifield (modified from Konishi 1991).

Chicken: Laminaris cells receive input from the ipsilateral nucleus magnocellularis (IPSI NM) onto their dorsal dendrites and input from the contralateral nucleus magnocellularis (CONTRA NM) onto their ventral dendrites. The ipsilateral inputs arrive simultaneously along the mediolateral extent of the nucleus laminaris, whereas the contralateral axons form delay lines, thus giving off collateral branches along the ventral surface of the nucleus. These delays are hypothesized to form a place map along the mediolateral axis of the nucleus. Delays from the ipsilateral and contralateral sides are approximately equal at the medial portion of the nucleus, and could map sound from the front of the bird. Because contralateral delays are longer than ipsilateral delays in the lateral portion of the nucleus, this region could map sounds that reached the contralateral ear first (modified from Overholt et al 1992).

cellular neurons of birds and reptiles. Bushy cells have large, round somata and thick axons, but also have many more dendrites than magnocellular neurons (Rhode et al 1983; Wu & Oertel 1984). Pioneering *in vitro* studies of the mouse cochlear nucleus show that bushy cells are well suited to

preserve the temporal firing pattern of auditory nerve inputs (Oertel 1985). They fire only one or two spikes in response to electrical stimulation of the auditory nerve and have nonlinear current-voltage relationships around the resting potential (inward rectification). Thus, when bushy cells are depolarized, their input resistance drops, thus causing rapid repolarization. The effects of excitation are brief and do not summate over time (Oertel 1985; Wu & Oertel 1984). Similar physiological responses may also characterize other phase-coding neurons. Bushy cells project to the medial superior olive in a circuit responsible for encoding and detecting interaural phase differences (see next section).

In addition to the circuit responsible for encoding and detecting interaural phase differences, a more mysterious structure has been found in the mammalian auditory system. In echolocating bats and dolphins, the columnar nucleus of the ventral lateral lemniscus (VNLLc) appears to be specialized for encoding temporal information (Covey & Casseday 1991). Its function is currently unknown, but its high degree of development and differentiation in two different mammalian orders suggest that this nucleus plays an important role in echolocation. The cells are innervated by the anteroventral cochlear nucleus and are physiologically suited to temporal coding, with little or no spontaneous activity, broad tuning, and short integration times (Covey & Casseday 1991).

To summarize, time coding is prominent in both the electrosensory and auditory systems. Both systems use similar algorithms with these major features: Phase-locked spikes encode the timing of the stimulus, and the CNS uses this code for the measurement of time disparities. Time and amplitude information is processed in parallel. Phase-coding improves with convergence from the periphery to higher level processing stations. The morphology and physiology of phase-coding cells is suited to their function; they have large cell bodies with few or no dendrites, thick axons, and short duration spikes.

DETECTING TIME DIFFERENCES

Behavioral experiments have shown that animals are capable of great accuracy in detecting time differences. How is this accomplished? The preferred model for detecting temporal disparities depends upon coincidence detection and proposes that a neuron that detects time differences responds best to simultaneous inputs. The model was first articulated by Jeffress (1948), with his place theory for detection of interaural time differences. The model circuit is composed of two elements: delay lines and coincidence detectors (Figure 2). The delay lines are created by varying axonal path lengths, and the coincidence detectors are neurons that

respond maximally when they receive simultaneous inputs, i.e. when the time difference is exactly compensated for by the delay introduced by the inputs.

The Jeffress model explains not only how interaural time differences are measured, but also how they are encoded. The circuit contains an array of coincidence detectors that receive input from afferent axons that serve as delay lines. Because of its position in the array, each neuron responds only to sound coming from a particular direction; thus, the anatomical place of the neuron encodes the location of the sound (Figure 2). These neurons use the temporal information to compute a new variable, time difference, and transform the time code into a place code. The selectivity of all higher-order auditory neurons to time difference derives from the "labeled-line" output of the place map (Konishi 1986).

Circuits in the fish midbrain, the auditory brainstem of barn owls, cats, and dogs, and the bat thalamus all contain neurons tuned to particular time delays between events. These delay-tuned ensembles employ similar algorithms. The circuits in the auditory brainstem resemble the Jeffress model in all its particulars, whereas the circuits in fish midbrain and bat thalamus differ from the Jeffress model in important respects. The circuit in the fish midbrain does not form a place map; the circuits in the bat thalamus do not contain axonal delay lines.

Detection of Phase Differences

When *Eigenmannia* is presented with sinusoidally varying electrical fields on two parts of its body, it can distinguish phase differences between these signals smaller than $1\ \mu\text{s}$. The circuit for detection of these phase differences is as follows: Phase-coding afferents from the medulla project to two cell types of the midbrain torus, synapsing on giant cell bodies and on the dendrites of the small cells (Figure 1). These afferents form local connections that encode the phase of the electric organ discharge from one part of the body surface. The giant cell's axons form horizontal connections that distribute this local phase information to small cells throughout the lamina, so that timing information from one part of the body surface may be compared with any other part. Small cells compare information from one patch of the body surface, through afferent input onto their dendrites, with phase information from any other part of the body surface, through the giant cell input to their cell bodies (Carr et al 1986b). Small cell responses encode either phase advance or phase delay (Heiligenberg & Rose 1985). The biophysical basis of the sensitivity to phase differences is as yet unknown, although a useful computer simulation has been generated (Lytton 1991). The small cell circuit allows the fish to perform all possible comparisons between different parts of the body surface, as required for

correct performance of the jamming avoidance response. Therefore, unlike the Jeffress model, the axonal delay lines do not create a place map of the computed stimulus feature, phase difference.

The phase sensitivity of the small cells is not as good as the behavior. Although the convergence of time-coding inputs progressively improves the temporal acuity of single neurons at each level of the time-coding pathway, the smallest temporal disparity detected by the most sensitive neurons in the torus is only 10 μ s, a value 20 times higher than that of the behavioral threshold (Carr et al 1986a). If the responses of such neurons are averaged over periods that are much longer than the 300 ms latency of the jamming avoidance response, a sensitivity to 1 μ s can be demonstrated (Rose & Heiligenberg 1985). Because the temporal averaging of the electrophysiologist is assumed to be equivalent to spatial averaging or parallel processing of the same signal in the CNS, this suggests that the sensitivity and short latency of the jamming avoidance response requires parallel convergence of large numbers of such toral neurons, which carry temporal information from wide areas of the body surface (Rose & Heiligenberg 1985). This convergence ultimately occurs at the level of single neurons in the prepacemaker nucleus (Kawasaki et al 1988).

To address the question of whether the sensitivity to small temporal disparities even exists at the single neuron level, Kawasaki et al (1988) recorded from neurons in the prepacemaker nucleus. Pacemaker neurons are located at the top of the sensory hierarchy that determines whether the jamming stimulus is higher or lower in frequency than the fish's own electric organ discharge. Pacemaker cells modulate their firing with changes in the interfering signal, even with temporal disparities as small as 1 μ s. According to Kawasaki et al (1988), it is unlikely that any neuron could discriminate such small temporal disparities in the arrival of two signals with certainty, but small differences in the timing of excitatory and inhibitory inputs could affect firing rates in a probabilistic fashion. Pooling the activity of many such neurons could yield predictable results (Kawasaki et al 1988). Thus, hyperacuity may be achieved by a network of pacemaker neurons.

Detection of Interaural Phase Differences in Birds and Mammals

The circuits in the auditory brainstem that detect interaural phase differences resemble the Jeffress model. The cochlear nucleus axons act as delay lines, and the laminaris or olivary neurons act as coincidence detectors, to form a circuit that measures and encodes interaural time differences (Figure 2). Physiological studies of the mammalian auditory system by

Goldberg & Brown (1969) and others found neuronal responses consistent with the Jeffress model. Recent results in the barn owl (Carr & Konishi 1990; Sullivan & Konishi 1986), cat (Smith et al 1990; Yin & Chan 1990), and chicken (Overholt et al 1992; Rubel & Parks 1975; Young & Rubel 1983) have described circuits that conform to the model's requirements. There are three major parts to the Jeffress model: delay lines, coincidence detection, and place coding of interaural phase difference. These are discussed in turn.

In the barn owl, magnocellular axons act as delay lines (Carr & Konishi 1988, 1990). They convey the timing or phase of the auditory stimulus in a bilateral projection to the nucleus laminaris, such that axons from the ipsilateral nucleus magnocellularis enter the nucleus laminaris from the dorsal side, whereas axons from the contralateral nucleus magnocellularis enter from the ventral side. Thus, these afferents interdigitate to innervate dorso-ventral arrays of neurons in laminaris in a sequential fashion (Figure 2). For each frequency band, recordings from these interdigitating ipsilateral and contralateral axons show regular changes in delay with depth in the nucleus laminaris (Carr & Konishi 1990). These conduction delays are similar to the 200 μ s range of interaural time differences available to the barn owl (Moiseff 1989). The conduction velocities of the delay lines are regulated precisely by the internodal distances within the nucleus laminaris. Normal internode lengths in the CNS are generally greater than 300 μ m (Waxman 1975), whereas internode lengths in the delay lines axons in the nucleus laminaris are both short and regular, occurring every 60 μ m (Carr & Konishi 1990).

The magnocellular-laminaris circuit in the chicken is similar to the circuit described for the barn owl, except that the nucleus laminaris is not a large nucleus, but a monolayer of bipolar cells oriented in the mediolateral dimension of the brainstem (Jhaveri & Morest 1982; Rubel & Parks 1975; Young & Rubel 1983) (Figure 2). Laminaris cells receive input from the ipsilateral nucleus magnocellularis onto their dorsal dendrites and input from the contralateral nucleus magnocellularis onto their ventral dendrites. The ipsilateral projections splay out to innervate laminaris neurons with approximately equal lengths to each cell, so that ipsilateral inputs arrive simultaneously along the mediolateral extent of the nucleus laminaris. Thus, the ipsilateral axons do not act as delay lines. The contralateral axons act as delay lines; each axon runs along the ventral surface of the nucleus laminaris, thus giving off collateral branches in the nucleus. Estimates of conduction velocity suggest that the contralateral delay line could encode 180 μ s of delay (Overholt et al 1992). Patterns of delay lines in the medial superior olive of the cat are very similar to those of the chicken. Axons from the contralateral cochlear nucleus form delay lines

across the rostro-caudal axis of the nucleus, whereas the ipsilateral axons form a less organized projection (Smith et al 1990).

The second major requirement of the Jeffress model is that the targets of the delay lines act as coincidence detectors. Goldberg & Brown (1969) showed that olivary neurons acted as coincidence detectors. The neurons of the avian nucleus laminaris and the mammalian medial superior olive phase-lock to both monaural and binaural stimuli and respond maximally when phase-locked spikes from each side arrive simultaneously, i.e. when the difference in the internal conduction delay is nullified by interaural time difference. The number of spikes elicited in response to a favorable interaural time difference is roughly double that elicited by a monaural stimulus; spike counts for unfavorable interaural time differences fall well below monaural response levels. Thus, physiological responses from these coincidence detectors are similar (Carr & Konishi 1990; Goldberg & Brown 1969; Sullivan & Konishi 1984; Yin & Chan 1990). The underlying biophysical mechanisms of coincidence detection are still unknown. Similarities between olivary cell responses and cross-correlated monaural spike trains have been used to suggest that cross-correlation provides a more general description of coincidence detection (Yin & Chan 1990); cross-correlation models for the computation of interaural phase differences are long standing in the auditory literature (Licklider 1959). The biophysical basis of cross-correlation is unclear, however, and more recent models do not depend upon cross-correlation (Colburn et al 1990; Grün et al 1990).

In one respect, owl laminaris neurons are different to all other cells that detect time differences. The delay line inputs are distributed evenly over the entire somato-dendritic area in the barn owl. In the chicken, laminaris neurons are bipolar, with inputs from the ipsilateral nucleus magno-cellularis restricted to the dorsal dendrites and contralateral inputs to the ventral dendrites of the laminaris neurons (Jhaveri & Morest 1982; Smith & Rubel 1979). In the mammalian medial superior olive, the principal cells are large bipolar ("rabbit ear") cells that segregate inputs from each ear onto the medial and lateral dendritic trees, respectively (Stotler 1953; Warr 1966). Even in the electric fish, the small cells segregate the inputs from the different parts of the body surface onto their soma and dendrite, respectively (Carr et al 1986b). Thus, owl laminaris neurons are the only "comparator" cells that do not segregate the inputs that are to be compared. The functional significance of these morphological differences is unknown.

Jeffress's model is called the place theory, because the array of coincidence detectors encode the computed variable of interaural phase difference by their place in the nucleus (see Figure 2). Each place in the nucleus laminaris or medial superior olive can be characterized by a pair of delays

for ipsi- and contralateral axons, and phase-locked spikes arrive at each neuron with delays unique to its place. The monaural delay lines, therefore, form neuronal maps of interaural time difference, or place, that are tapped by coincidence detectors. Evidence for a place map was obtained from the medial superior olive of the cat (Yin & Chan 1990). Cells in the medial superior olive respond to a range of interaural time differences between 0 and 400 μ s, thus corresponding to locations in the contralateral sound field. The locations of recordings were marked with lesions; best interaural time differences near 0 μ s were located at the anterior pole of the olive, whereas larger time differences were located at more posterior locations. Thus, the medial superior olive appears to contain a map of place in each frequency band. The nucleus laminaris in the chicken may also contain maps of interaural time difference or place organized along the medio-lateral dimension of the nucleus, with best interaural time differences of 0° close to the medial edge (Figure 2) (Overholt et al 1992). In the barn owl, the nucleus laminaris contains many maps of interaural time difference or place within each frequency band, repeated along the dorso-ventral dimension of the nucleus (Figure 2). In these maps of interaural time difference, 0° is represented near the ventral edge of the nucleus (Sullivan & Konishi 1986). Thus, the barn owl appears to have duplicated the ancestral avian pattern and rotated the place map by 90° (Takahashi & Konishi 1988). These duplicated maps may confer increased sensitivity to interaural time differences on this nocturnal predator.

Bats and the Detection of Echo Delays

A bat emits orientation sounds and listens to the returning echos. The delay between the pulse and echo conveys the target distance. For example, a 1 ms echo delay corresponds to a 17.3 cm target distance (Suga 1990). How are these delays encoded and processed in the CNS? At the periphery, neurons respond to both the pulse and the echo, and the time interval between the two grouped discharges is directly related to the echo delay. The medial geniculate body and auditory cortex, however, contain neurons that respond poorly or not at all to the echo alone, but respond strongly when pulse and echo are combined with a particular echo delay. These “combination-sensitive” neurons are called FM-FM cells, and were first found in the auditory cortex of the mustache bat (O'Neill & Suga 1982; Suga et al 1978). Tuning to echo-delay is mapped on the cortical surface to form maps of target range (O'Neill & Suga 1982). To create an FM-FM cell, the bat must differentiate between pulse and echo. Sonar pulses contain four harmonics, and the bat uses the low amplitude first harmonic (FM₁) to mark the pulse and the other harmonics (FM₂, FM₃, FM₄) to mark the echo.

Where are these FM_1 - FM_n neurons created? There are no FM_1 - FM_n neurons in the inferior colliculus (O'Neill 1985), and delays between pulse- and echo-evoked activity may still be found in the brachium of the inferior colliculus (Kuwabara & Suga 1988). FM_1 - FM_n neurons are first found in the medial geniculate body (Olsen & Suga 1991). FM_1 and FM_n channels from the inferior colliculus converge in the medial geniculate body to create FM_1 - FM_n combination sensitive neurons (Figure 3). For the FM_1 and FM_n evoked activity to coincide in the medial geniculate body, the FM_1 signal must be delayed with reference to the FM_n . How are the FM_1 delay lines created? The answer is more complex than that from the auditory brainstem because of the time delays involved. The pulse-echo delay axis is from 0.4 to 18 ms, i.e. a range axis of 7–310 cm (Suga 1990). These delays are much larger than the 200 μ s interaural time differences available to the barn owl (Carr & Konishi 1990) or 400 μ s for the cat (Yin & Chan 1990), and it would be tortuous to create them by varying axonal path lengths. Olsen & Suga (1991) have found evidence for at least two other mechanisms for delaying the FM_1 signal. For best delays below 4 ms, delay tuning depends only on coincidence of pulse and echo excitation.

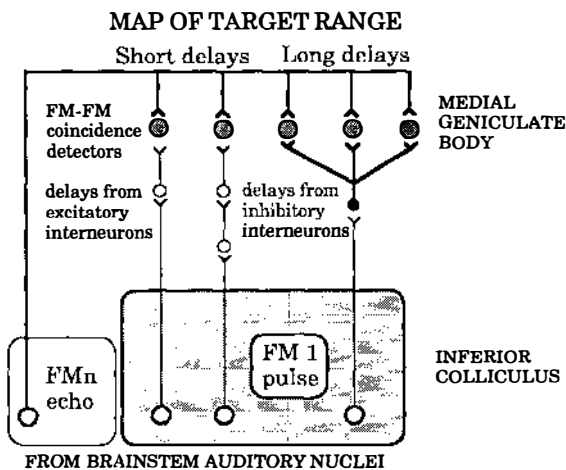


Figure 3 Schematic circuit for creation of delay-tuned FM-FM cells in the medial geniculate body of the thalamus. Auditory information about the pulse (FM_1) and echo (FM_n) travels in different frequency channels from the auditory brainstem to the inferior colliculus, and from there to the medial geniculate body. The neurons tuned to the echo are not delayed in the medial geniculate, whereas neurons tuned to the pulse exhibit both short and long delays. For short delays (0.5–4 ms), the pulse signal is delayed by excitatory relay interneurons. Long delays (4–18 ms) appear to be created by inhibitory interneurons (modified from Suga 1992).

This result is consistent with delay lines created by excitatory relay units (Figure 3). For long best delays (>4 ms), the FM_1 pulse produces inhibition of short latency and variable duration, followed by excitation, consistent with the "inhibitory gate" hypothesis (Sullivan 1982), in which FM_1 delays are created by varying the duration of inhibition. The inhibition may be produced by local circuits within the medial geniculate body (Olsen & Suga 1991).

These inhibitory local circuits in the auditory thalamus have a parallel in the visual thalamus. In the lateral geniculate, inhibition produces the X-lagged cell responses (Mastronade 1987a,b). These geniculate cells differ from their retinal X-afferent inputs and from adjacent nonlagged geniculate X-cells in displaying an early inhibition rather than excitation to the onset of a visual stimulus in their receptive field. The X-lagged response is delayed, because the excitatory input is blocked by inhibition, not because excitation is delayed. Postinhibitory rebound then produces the lagged or phase-shifted response. The response timing differences between lagged and nonlagged cells may be important for calculating direction selectivity in visual cortex (Saul & Humphrey 1990).

CONCLUSIONS

Analysis of the encoding and processing of temporal information has uncovered some common organizational principles. Time-coding systems implement similar algorithms for the encoding and processing of temporal information (Konishi 1991). Despite different neural substrates, CNS time channels share numerous morphological and physiological adaptations to improve the time coding of signal. In the electric sense and audition, a dichotomy exists between coding for the timing or phase of the signal and coding for its amplitude. In electric fish, the dichotomy between phase and amplitude coding begins at the receptor level; in the auditory system, the separation into phase and amplitude coding is derived within the CNS. A similar separation may be found in other sensory systems; visual information is processed in two channels that differ in their spatial and temporal acuity (Livingstone & Hubel 1987; Maunsell & Newsome 1987).

Time-coding systems also implement similar algorithms for the measurement of time differences. These algorithms all depend on delay lines and coincidence detectors in some form. Short delays may be provided by axonal delay lines. Axons can provide functionally significant delays in conduction. Compensatory axonal delay line mechanisms include equalization of path length, differences in conduction velocity, and localized delays, determined by variations in preterminal axon branches (Waxman 1975). Long delays may be introduced by interposed excitatory synapses, whereas

the longest delays originate with inhibitory interneurons and regulation of post-inhibitory rebound. Neural circuits that incorporate delays may be widespread in the CNS. Such circuits are also found in many neural models; for example, early models of directional selectivity in the visual system connected two detectors to an AND-NOT gate, one via a delay (reviewed in Marr 1982). The underlying mechanisms of coincidence detection are less well understood than those that generate delays, and the biophysical basis of coincidence detection represents an important challenge. Coincidence detectors transform timing information into a place code for further processing and the formation of perceptions, such as sound location or target range.

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Literature Cited

- Altes, R. A. 1989. Ubiquity of hyperacuity. *J. Acoust. Soc. Am.* 85: 943–52.
- Bell, C. C., Zakon, H., Finger, T. E. 1989. Mormyromast electroreceptor organs and their afferent fibers in mormyrid fish: I. Morphology. *J. Comp. Neurol.* 286: 391–409.
- Bennett, M. V. L. 1970. Comparative physiology: electric organs. *Annu. Rev. Physiol.* 32: 471–531.
- Boord, R. L. 1968. Ascending projections of the primary cochlear nuclei and nucleus laminaris in the pigeon. *J. Comp. Neurol.* 133: 523–42.
- Brawer, J. R., Morest, D. K. 1974. Relations between auditory nerve endings and cell types in the cat's anteroventral cochlear nucleus seen with Golgi method and Nomarski optics. *J. Comp. Neurol.* 160: 491–506.
- Bullock, T. H., Hamstra, R. H., Scheich, H. 1972. The jamming avoidance response of high frequency electric fish. I. General features. *J. Comp. Physiol.* 77: 1–22.
- Carr, C. E. 1992. The evolution of the central auditory system in reptiles and birds. In *The Evolutionary Biology of Hearing*, ed. D. B. Webster, R. R. Fay, A. N. Popper, pp. 511–44. New York: Springer-Verlag.
- Carr, C. E., Boudreau, R. E. 1992. Organization of the nucleus magnocellularis and the nucleus laminaris in the barn owl: encoding and measuring interaural time differences. *J. Comp. Neurol.* In press.
- Carr, C., Heiligenberg, W., Rose, G. 1986a. A time-comparison circuit in the electric fish midbrain. I. Behavior and physiology. *J. Neurosci.* 6: 107–19.
- Carr, C. E., Konishi, M. 1990. A circuit for detection of interaural time differences in the brainstem of the barn owl. *J. Neurosci.* 10: 3227–46.
- Carr, C. E., Konishi, M. 1988. Axonal delay lines for time measurement in the owl's brainstem. *Proc. Natl. Acad. Sci. USA* 85: 8311–15.
- Carr, C. E., Maler, L., Taylor, B. 1986b. A time comparison circuit in the electric fish midbrain. II. Functional morphology. *J. Neurosci.* 6: 1372–83.
- Colburn, H. S., Han, Y., Culotta, C. P. 1990. Coincidence model of MSO responses. *Hear. Res.* 49: 335–55.
- Covey, E., Casseday, J. H. 1991. The monaural nuclei of the lateral lemniscus in an echolocating bat: parallel pathways for analyzing temporal features of sound. *J. Neurosci.* 11: 3456–70.

- Fay, R. R. 1988. *Hearing in Vertebrates: A Psychophysics Databook*. Winnetka, Ill: Hill-Fay Assoc.
- Goldberg, J. M., Brown, P. B. 1969. Response of binaural neurons of dog superior olivary complex to dichotic tonal stimuli: Some physiological mechanisms of sound localization. *J. Neurophysiol.* 32: 613–36
- Grün, S., Aertsen, A., Wagner, H., Carr, C. 1990. Sound localization in the barn owl: a quantitative model of binaural interaction in the nucleus laminaris. *Soc. Neurosci. Abstr.* 16
- Heffner, R. S., Heffner, H. E. 1992. Evolution of sound localization in mammals. In *The Evolutionary Biology of Hearing*, ed. D. B. Webster, R. R. Fay, A. N. Popper, pp. 691–716. New York: Springer-Verlag
- Heiligenberg, W. 1991. *Neural Nets in Electric Fish*. Cambridge, Mass: MIT Press
- Heiligenberg, W., Baker, C., Matsubara, J. A. 1978. The jamming avoidance response in Eigenmannia revisited: The structure of a neuronal democracy. *J. Comp. Physiol.* 127: 267–86
- Heiligenberg, W., Rose, G. 1985. Phase and amplitude computations in the midbrain of an electric fish: intracellular studies of neurons participating in the jamming avoidance response of Eigenmannia. *J. Neurosci.* 5: 515–31
- Hopkins, C. D. 1986. Temporal structure of non-propagated electric communication signals. *Brain Behav. Evol.* 31: 43–59
- Jeffress, L. A. 1948. A place theory of sound localization. *J. Comp. Physiol. Psychol.* 41: 35–39
- Jhaveri, S., Morest, K. 1982. Neuronal architecture in nucleus magnocellularis of the chicken auditory system with observations on nucleus laminaris: A light and electron microscope study. *Neuroscience* 7: 809–36
- Kawasaki, M., Rose, G., Heiligenberg, W. 1988. Temporal hyperacuity in single neurons of electric fish. *Nature* 336: 173–76
- Kiang, N. Y. S., Watanabe, T., Thomas, E. C., Clark, E. F. 1965. *Discharge Patterns of Single Fibers in the Cat's Auditory Nerve*. Cambridge, Mass: MIT Press
- Kidd, R. C., Weiss, T. F. 1990. Mechanisms that degrade timing information in the cochlea. *Hear. Res.* 49: 181–208
- Knudsen, E. I., Konishi, M. 1979. Sound localization by the barn owl (Tyto alba). *J. Comp. Physiol.* 133: 1–11
- Konishi, M. 1991. Deciphering the brain's codes. *Neural Comput.* 3: 1–18
- Konishi, M. 1986. Centrally synthesized maps of sensory space. *Trends Neurosci.* 9: 163–68
- Konishi, M. 1973. How the owl tracks its prey. *Am. Sci.* 61: 414–24
- Kuwabara, N., Suga, N. 1988. Mechanism for production of “range-tuned” neurons in the mustached bat: delay lines and amplitude selectivity are created by the midbrain auditory nuclei. *Assoc. Res. Otolaryngol. Abstr.* 11: 200
- Licklider, J. C. R. 1959. Three auditory theories. In *Psychology: A Study of a Science*, ed. S. Koch, pp. 41–144. New York: McGraw-Hill
- Livingstone, M. L., Hubel, D. H. 1987. Psychophysical evidence for separate channels for the perception of form, color, movement and depth. *J. Neurosci.* 7: 3416–68
- Lytton, W. W. 1991. Simulations of a phase comparing neuron of the electric fish Eigenmannia. *J. Comp. Physiol.* 169: 117–25
- Maler, L., Sas, E., Rogers, J. 1981. The cytology of the posterior lateral line lobe of high frequency weakly electric fish (Gymnotoidei): Dendritic differentiation and synaptic specificity in a simple cortex. *J. Comp. Neurol.* 195: 87–140
- Marr, D. 1982. *Vision*. New York: Freeman
- Masters, W. M., Moffat, A. J. M., Simmons, J. A. 1985. Sonar tracking of horizontally moving targets by the big brown bat *Eptesicus fuscus*. *Science* 228: 1331–33
- Mastronade, D. N. 1987a. Two classes of single-input X-cells in cat lateral geniculate nucleus. I. Receptive field properties and classification of cells. *J. Neurophysiol.* 57: 357–80
- Mastronade, D. N. 1987b. Two classes of single input cell in the lateral geniculate nucleus. II. Retinal inputs and the generation of receptive field properties. *J. Neurophysiol.* 57: 381–413
- Maunsell, J. H. R., Newsome, W. T. 1987. Visual processing in monkey extrastriate cortex. *Annu. Rev. Neurosci.* 10: 363–402
- Mills, A. W. 1972. Auditory localization. In *Foundations of Modern Auditory Theory*, ed. J. V. Tobias, 2: 303–48. New York: Academic
- Moiseff, A. 1989. Binaural disparity cues available to the barn owl for sound localization. *J. Comp. Physiol.* 164: 629–36
- Moiseff, A., Konishi, M. 1981. Neuronal and behavioral sensitivity to binaural time differences in the owl. *J. Neurosci.* 1: 40–48
- Moss, C. F., Schnitzler, H.-U. 1989. Accuracy of target ranging in echolocating bats: acoustic information processing. *J. Comp. Physiol.* 165: 383–93
- Moss, C. F., Simmons, J. A. 1992. Acoustic image representation of a point target in the FM-FM bat, *Eptesicus fuscus*: Evi-

- dence for the perception of echo phase in sonar. *J. Acoust. Soc. Am.* In press
- O'Neill, W. E. 1985. Responses to pure tones and linear FM components of the CF-FM biosonar signal by single units in the inferior echolocating of the mustached bat. *J. Comp. Physiol.* 157: 797-815
- O'Neill, W. E., Suga, N. 1982. Encoding of target range information and its representation in the auditory cortex of the mustache bat. *J. Neurosci.* 2: 17-31
- Oertel, D. 1985. Use of brain slices in the study of the auditory system: Spatial and temporal summation of synaptic inputs in cells in the anteroventral cochlear nucleus of the mouse. *J. Acoust. Soc. Am.* 78: 328-33
- Olsen, J. F., Suga, N. 1991. Combination-sensitive neurons in the medial geniculate body of the mustached bat: Encoding of target range information. *J. Neurophysiol.* 65: 1275
- Overholt, T., Hyson, R., Rubel, E. W. 1992. A delay-line circuit for coding interaural time differences in the chick brain stem. *J. Neurosci.* 12: 1698-1708
- Payne, R. S. 1971. Acoustic localization of prey by barn owls (*Tyto alba*). *J. Exp. Biol.* 54: 535-73
- Rhode, W. S., Oertel, D., Smith, P. H. 1983. Physiological response properties of cells labeled intracellularly with horseradish peroxidase in cat ventral cochlear nucleus. *J. Comp. Neurol.* 213: 448-63
- Rose, G., Heiligenberg, W. 1985. Temporal hyperacuity in the electric sense of fish. *Nature* 318: 178-80
- Rubel, E. W., Parks, T. N. 1975. Organization and development of brainstem auditory nuclei of the chicken: Tonotopic organization of *N. magnocellularis* and *N. laminaris*. *J. Comp. Neurol.* 164: 411-34
- Ryugo, D. K., Fekete, D. M. 1982. Morphology of primary axosomatic endings in the anteroventral cochlear nucleus of the cat: A study of the endbulbs of Held. *J. Comp. Neurol.* 210: 239-57
- Saul, A. B., Humphrey, A. L. 1990. Spatial and temporal response properties of lagged and non-lagged cells in cat lateral geniculate nucleus. *J. Neurophysiol.* 64: 206-24
- Scheich, H., Bullock, T. H., Hamstra, R. H. 1973. Coding properties of two classes of afferent nerve fibers: High frequency electroreceptors in the electric fish *Eigenmannia*. *J. Neurophysiol.* 36: 39-60
- Simmons, J. A. 1979. Perception of echo phase information in bat sonar. *Science* 204: 1336-38
- Simmons, J. A. 1973. The resolution of target range by echolocating bats. *J. Acoust. Soc. Am.* 54: 157-73
- Simmons, J. A., Ferragamo, M., Moss, C. F., Stevenson, S. B., Altes, R. A. 1990. Discrimination of jittered sonar echos by the echolocating bat, *Eptesicus fuscus*: The shape of target images in echolocation. *J. Comp. Physiol.* 167: 589-616
- Smith, P. H., Joris, P. X., Yin, T. C. T. 1990. Projections of spherical bushy cells to the MSO in the cat: evidence for delay lines. *Neurosci. Abstr.* 16: 723
- Smith, Z. D. J., Rubel, E. W. 1979. Organization and development of brainstem auditory nuclei of the chicken: Dendritic gradients in nucleus laminaris. *J. Comp. Neurol.* 186: 213-39
- Spzir, M. R., Sento, S., Ryugo, D. K. 1990. The central projections of the cochlear nerve fibers in the alligator lizard. *J. Comp. Neurol.* 295: 530-47
- Stotler, W. A. 1953. An experimental study of the cells and connections of the superior olivary complex of the cat. *J. Comp. Neurol.* 98: 401-32
- Suga, N. 1992. Parallel-hierarchical processing of complex sounds for specialized auditory function. In *Handbook of Acoustics*, ed. M. J. Crocker. New York: Wiley
- Suga, N. 1990. Cortical computational maps for auditory imaging. *Neural Netw.* 3: 3-21
- Suga, N., O'Neill, W. E., Manabe, T. 1978. Cortical neurons sensitive to combinations of information bearing elements of biosonar signals in the mustache bat. *Science* 200: 778-81
- Sullivan, W. E. 1982. Possible neural mechanisms of target distance coding in auditory system of the echolocating bat *Myotis lucifugus*. *J. Neurophysiol.* 48: 1033-47
- Sullivan, W. E., Konishi, M. 1986. Neural map of interaural phase difference in the owl's brainstem. *Proc. Natl. Acad. Sci. USA* 83: 8400-4
- Sullivan, W. E., Konishi, M. 1984. Segregation of stimulus phase and intensity coding in the cochlear nucleus of the barn owl. *J. Neurosci.* 4: 1787-99
- Szabo, T. 1965. Sense organs of the lateral line system in some electric fish of the Gymnotidae, Mormyridae and Gymnarchidae. *J. Morphol.* 117: 229-50
- Takahashi, T., Konishi, M. 1988. The projections of the cochlear nuclei and nucleus laminaris to the inferior colliculus of the barn owl. *J. Comp. Neurol.* 274: 190-211
- Takahashi, T., Moiseff, A., Konishi, M. 1984. Time and intensity cues are processed independently in the auditory system of the owl. *J. Neurosci.* 4: 1781-86
- Von der Emde, G., Bleckmann, H. 1991. Extreme phase sensitivity of afferents which innervate mormyromast electroreceptors. *Naturwissenschaften* 78: 131-33

- Warr, W. B. 1966. Fiber degeneration following lesions in the anterior ventral cochlear nucleus of the cat. *Exp. Neurol.* 14: 453–74
- Waxman, S. G. 1975. Integrative properties and design principles of axons. *Int. Rev. Neurobiol.* 18: 1–40
- Weiss, T. F., Rose, C. 1988. A comparison of synchronization filters in different auditory receptor organs. *Hear. Res.* 33: 175–80
- Wu, S. H., Oertel, D. 1984. Intracellular injection with horseradish peroxidase of physiologically characterized stellate and bushy cells in slices of mouse anteroventral cochlear nucleus. *J. Neurosci.* 4: 1577–88
- Yin, T. C. T., Chan, J. C. K. 1990. Interaural time sensitivity in medial superior olive of cat. *J. Neurophysiol.* 64: 465–88
- Young, S. R., Rubel, E. W. 1983. Frequency-specific projections of individual neurons in chick brainstem auditory nuclei. *J. Neurosci.* 7: 1373–78